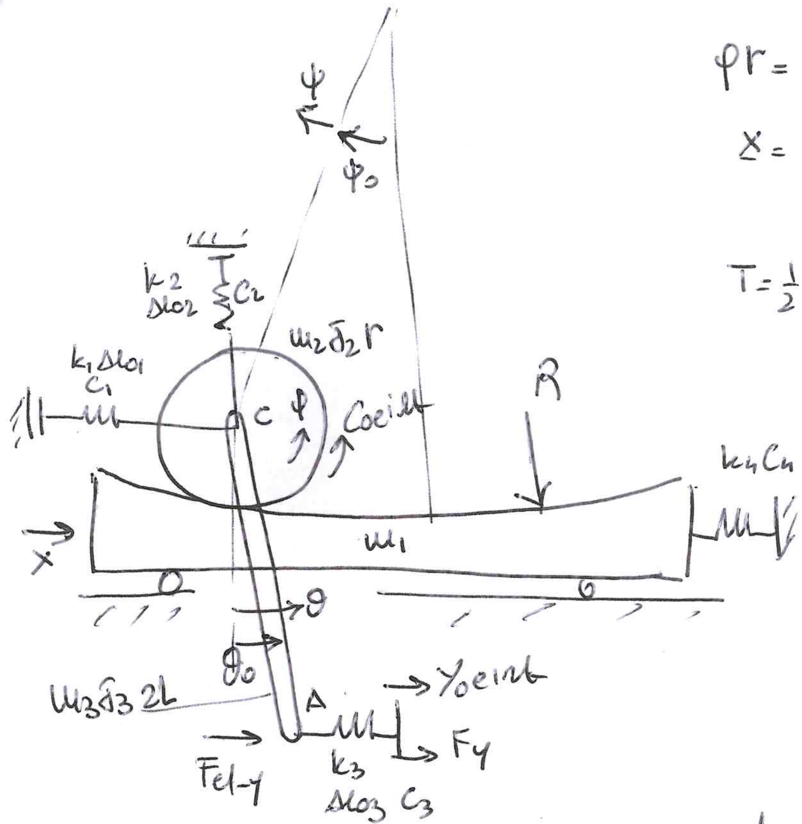


$$\varphi r = (R-r)\psi \Rightarrow \psi = \frac{r}{R-r}\varphi = \eta\varphi$$

$$\underline{x} = \begin{Bmatrix} x \\ \varphi \\ \theta \end{Bmatrix}$$

$$T = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\dot{x}_c^2 + \frac{1}{2}m_2\dot{y}_c^2 + \frac{1}{2}J_2\dot{\varphi}^2 + \frac{1}{2}m_3\dot{x}_G^2 + \frac{1}{2}m_3\dot{y}_G^2 + \frac{1}{2}J_3\dot{\theta}^2$$

$$V_{EL} = \frac{1}{2}\sum_{i=1}^{4} k_i\Delta l_i^2 \; ; \; D = \frac{1}{2}\sum_{i=1}^{4} c_i\Delta \dot{l}_i^2$$

$$\delta W = C_0 e^{i\omega t}\delta\varphi + F_{el-y}\,\delta x_A$$

$$\begin{cases} x_c = x - (R-r)\,\mathrm{sen}(\psi_0 + \eta\varphi) \\ y_c = -(R-r)\cos(\psi_0 + \eta\varphi) \\ x_G = x - (R-r)\,\mathrm{sen}(\psi_0 + \eta\varphi) + L\,\mathrm{sen}\,\theta \\ y_G = -(R-r)\cos(\psi_0 + \eta\varphi) - L\cos\theta \\ x_A = x - (R-r)\,\mathrm{sen}(\psi_0 + \eta\varphi) + 2L\,\mathrm{sen}\,\theta \end{cases}$$

$$\begin{cases} \Delta l_1 = x_c - x_{c0} \\ \Delta l_2 = -y_c + y_{c0} \\ \Delta l_3 = -x_A + x_{A0} \\ \Delta l_4 = -x \end{cases}$$

$$\begin{cases} \frac{\partial x_c}{\partial x} = 1 \; ; \; \frac{\partial x_c}{\partial \varphi} = -r\cos(\psi_0 + \eta\varphi) \; ; \; \frac{\partial x_c}{\partial \theta} = 0 \\ \frac{\partial y_c}{\partial x} = 0 \; ; \; \frac{\partial y_c}{\partial \varphi} = +r\,\mathrm{sen}(\psi_0 + \eta\varphi) \; ; \; \frac{\partial y_c}{\partial \theta} = 0 \\ \frac{\partial x_G}{\partial x} = 1 \; ; \; \frac{\partial x_G}{\partial \varphi} = -r\cos(\psi_0 + \eta\varphi) \; ; \; \frac{\partial x_G}{\partial \theta} = L\cos\theta \\ \frac{\partial y_G}{\partial x} = 0 \; ; \; \frac{\partial y_G}{\partial \varphi} = +r\,\mathrm{sen}(\psi_0 + \eta\varphi) \; ; \; \frac{\partial y_G}{\partial \theta} = +L\,\mathrm{sen}\,\theta \\ \frac{\partial x_A}{\partial x} = 1 \; ; \; \frac{\partial x_A}{\partial \varphi} = -r\cos(\psi_0 + \eta\varphi) \; ; \; \frac{\partial x_A}{\partial \theta} = 2L\cos\theta \end{cases}$$

$[M_{PH}] = diag\,(m_1 \;\; m_2 \;\; m_2 \;\; J_2 \;\; m_3 \;\; m_3 \;\; J_3)$

$[C_{PH}] = diag\,(c_1 \;\; c_2 \;\; c_3 \;\; c_4) \;;\; [k_{PH}] = diag\,(k_1 \;\; k_2 \;\; k_3 \;\; k_4)$

$\underline{F} = \{ C_{oe} \, i\omega t \;\; F_{el\_y} \}^T$

$$[\Lambda_m] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -r\cos\varphi_0 & 0 \\ 0 & r\,sen\,\varphi_0 & 0 \\ 0 & \cancel{}\,1 & 0 \\ 1 & -r\cos\varphi_0 & L\cos\vartheta_0 \\ 0 & r\,sen\,\varphi_0 & L\,sen\,\vartheta_0 \\ \cancel{1}\,0 & \cancel{-r\cos\varphi_0}\;0 & \cancel{L\cos\vartheta_0}\;1 \end{bmatrix} \qquad [\Lambda_k] = \begin{bmatrix} 1 & -r\cos\varphi_0 & 0 \\ 0 & -r\,sen\,\varphi_0 & 0 \\ -1 & r\cos\varphi_0 & -2L\cos\vartheta_0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$[\Lambda_e] = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -r\cos\varphi_0 & 2L\cos\vartheta_0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta l_1}{\partial \varphi^2} = \frac{r^2}{R-r}\,sen\,\varphi_0 \;\rightarrow\; [k_{II}\,el_1] = k_1 \Delta l_{01} \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{r^2}{R-r}\,sen\,\varphi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta l_2}{\partial \varphi^2} = -\frac{r^2}{R-r}\cos\varphi_0 \;\rightarrow\; [k_{II}\,el_2] = k_2 \Delta l_{02} \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{-r^2}{R-r}\cos\varphi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta l_3}{\partial \varphi^2} = -\frac{r^2}{R-r}\,sen\,\varphi_0$$

$$\frac{\partial^2 \Delta l_3}{\partial \vartheta^2} = 2L\,sen\,\vartheta_0 \;\rightarrow\; [k_{II}, el_3] = k_3 \Delta l_{03} \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{-r^2}{R-r}\,sen\,\varphi_0 & 0 \\ 0 & 0 & 2L\,sen\,\vartheta_0 \end{bmatrix}$$

$h_{g2} = Y_c$

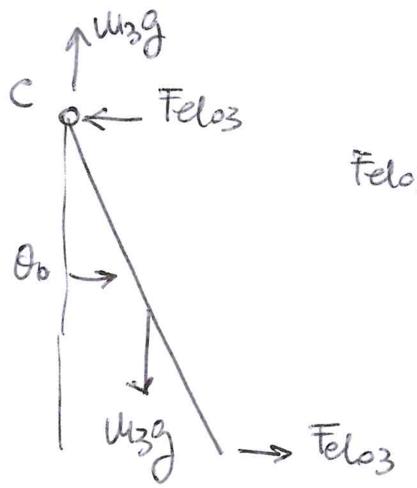
$$\frac{\partial^2 Y_c}{\partial \varphi^2} = \frac{r^2}{R-r}\cos\varphi_0 \;\rightarrow\; [k_{g2}] = m_2 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{r^2}{R-r}\cos\varphi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$h_{43} = y_4$

$$\frac{\partial^2 h_{43}}{\partial \varphi^2} = \frac{r^2}{R-r}\cos\psi_0$$

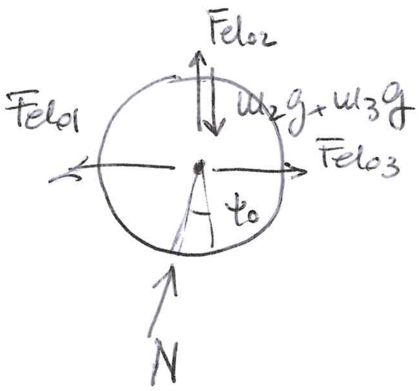
$$\frac{\partial^2 h_{43}}{\partial \theta^2} = L\cos\vartheta_0$$

$$[k_{43}] = m_3 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{r^2}{R-r}\cos\psi_0 & 0 \\ 0 & 0 & L\cos\vartheta_0 \end{bmatrix}$$



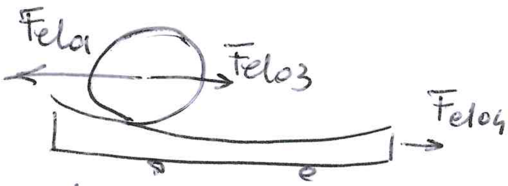
$$F_{el03}\, 2\cancel{L}\cos\vartheta_0 = m_3 g\, \cancel{L}\, \text{sen}\,\vartheta_0$$

$$\Delta l_{03} = \frac{m_3 g}{2k_3}\,\text{Tg}\,\vartheta_0$$

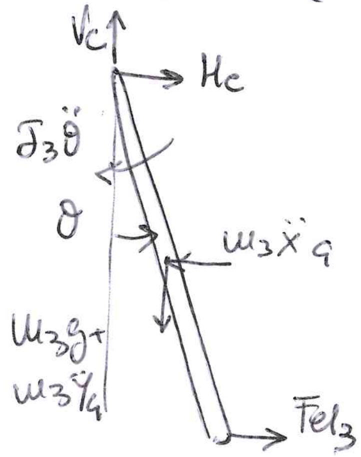


$$(F_{el01} - F_{el03})\,\cancel{r}\cos\psi_0 = (m_2 g + m_3 g - F_{el02})\,\cancel{r}\,\text{sen}\,\psi_0$$

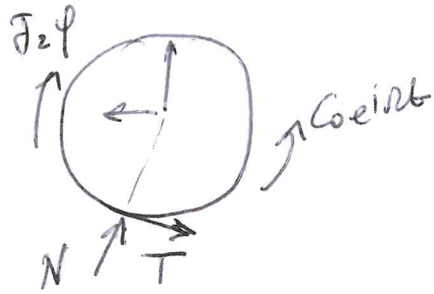
$$\Delta l_{02} = \left[m_2 g + m_3 g - \frac{F_{el01} - F_{el03}}{\text{Tg}\,\psi_0}\right]\frac{1}{k_2}$$



$$F_{el04} = F_{el01} - F_{el03}$$



$$H_c = m_3\ddot{x}_4 - F_{el3}$$



$$T = \frac{J_2\ddot{\varphi} - C_0 e^{i\omega t}}{r}$$

$$X = \begin{Bmatrix} x \\ \varphi \\ \theta \end{Bmatrix}$$

$$\Lambda_{du} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -(R-r)\cos\varphi_0 & 0 \\ 0 & (R-r)\,\text{sen}\,\varphi_0 & 0 \\ 0 & \frac{R-r}{r} & 0 \\ 1 & -(R-r)\cos\varphi_0 & L\cos\vartheta_0 \\ 0 & (R-r)\,\text{sen}\,\varphi_0 & L\,\text{sen}\,\vartheta_0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[\Lambda_{dk}] = \begin{bmatrix} 1 & -(R-r)\cos\varphi_0 & 0 \\ 0 & -(R-r)\,\text{sen}\,\varphi_0 & 0 \\ -1 & (R-r)\cos\varphi_0 & -2L\cos\vartheta_0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$[\Lambda_{de}] = \begin{bmatrix} 0 & \frac{R-r}{r} & 0 \\ 1 & -(R-r)\cos\varphi_0 & 2L\cos\vartheta_0 \end{bmatrix}$$

$$[k_{\text{II}el_1}] = k_1\Delta l_{01} \begin{bmatrix} 0 & 0 & 0 \\ 0 & (R-r)\,\text{sen}\,\varphi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (k_{\text{II}el_2}) = k_2\Delta l_{02} \begin{bmatrix} 0 & 0 & 0 \\ 0 & -(R-r)\cos\varphi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(k_{\text{II}el_3}) = k_3\Delta l_{03} \begin{bmatrix} 0 & 0 & 0 \\ 0 & -(R-r)\,\text{sen}\,\varphi_0 & 0 \\ 0 & 0 & 2L\,\text{sen}\,\vartheta_0 \end{bmatrix}$$

$$[k_{g2}] = m_2 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & (R-r)\cos\varphi_0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [k_{g3}] = m_3 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & (R-r)\cos\varphi_0 & 0 \\ 0 & 0 & L\cos\vartheta_0 \end{bmatrix}$$